Paper e-Brochure

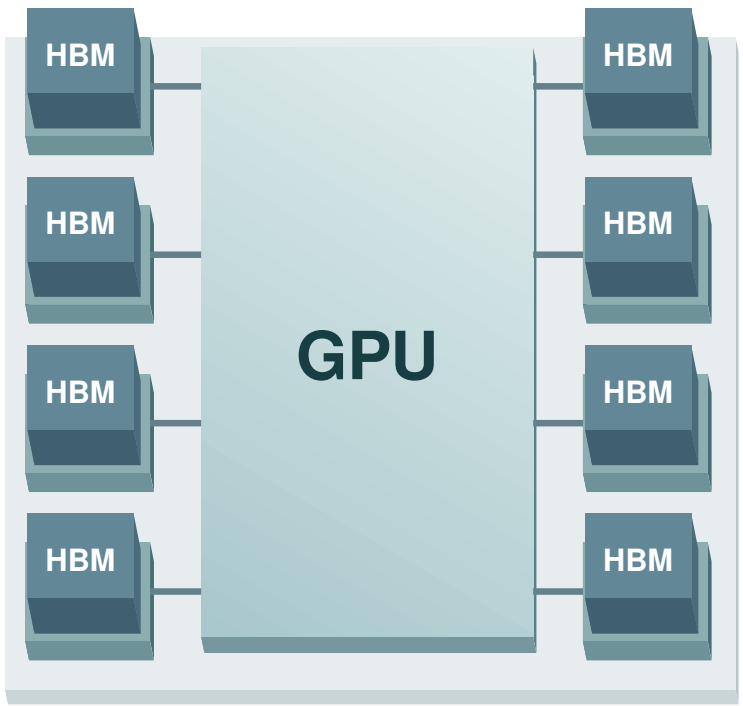
Custom-HBM Array

LLM Decode Accelerator

The decoding bottleneck

Memory-bound LLM decoding leaves traditional GPU compute underutilized, while HBM bandwidth is difficult to scale.

Conventional GPU + HBM

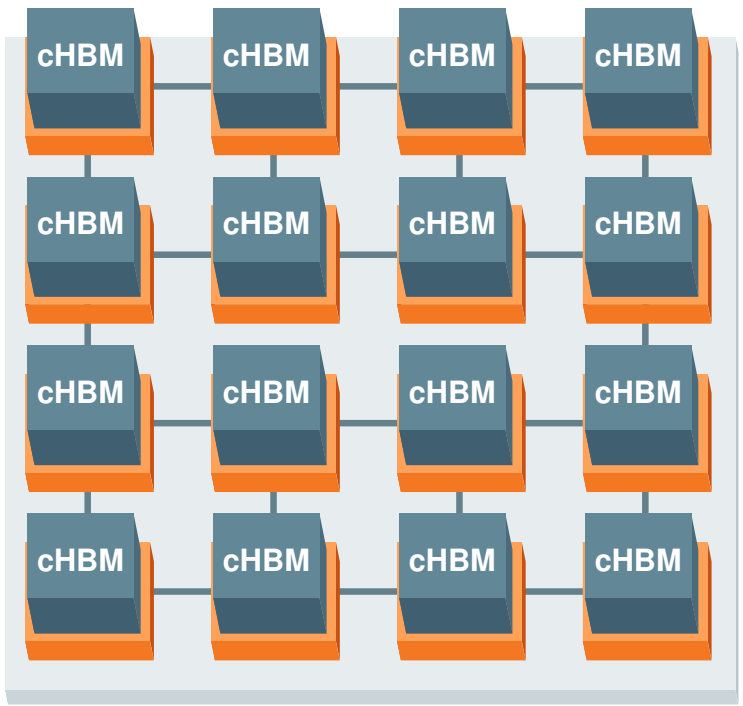


High memory use, low compute utilization

Scalable cHBM chiplets

Each cHBM chiplet integrates an NPU into its logic die for LLM decoding. The array scales memory bandwidth to match workload needs.

4 × 4 cHBM Array

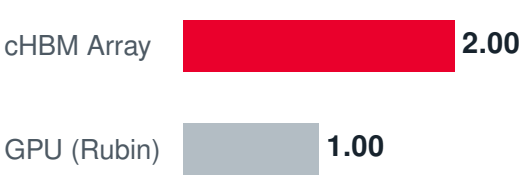


Hybrid tensor and pipeline parallelism

Performance

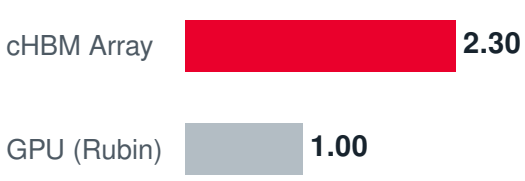
2.00×
Memory bandwidth

Memory bandwidth



UP TO **2.30×**
Decode throughput

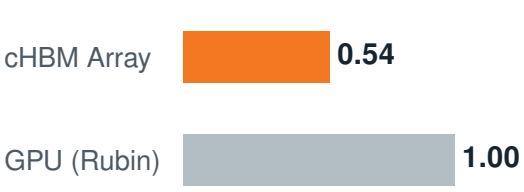
Tokens per second



Energy efficiency

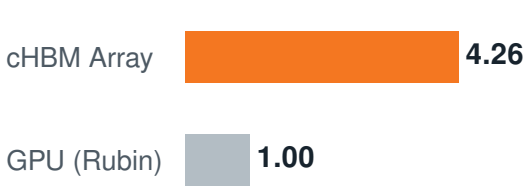
–46%
Power consumption

Power consumption



UP TO **4.26×**
Performance per watt

Tokens per second / watt



Relative to GPU (Rubin). Chart values normalize GPU (Rubin) to 1.00.